CE7455: Deep Learning for Natural Language Processing

Final Project Proposal

Extraction of Social and Clinical Impacts of Substance Use from Social Media Posts

Group ID: [G26]

Team Members: ISMAIL MAMDOUH MOSAAD ELYAMANY [ISMAILMA001@e.ntu.edu.sg]
HUANG CHONGTIAN [CHONGTIA002@e.ntu.edu.sg]
XIA YITONG [YITONG010@e.ntu.edu.sg]
ZHU RUIJIE [RUIJIE008@e.ntu.edu.sg]

1. Goals and Objectives

Singapore maintains one of the world's strictest drug enforcement regimes, yet substance use and its consequences — job loss, family breakdown, mental health deterioration — remain underreported. People affected rarely disclose these experiences to authorities or healthcare providers. They do, however, write about them online. The same pattern holds globally: Reddit communities around substance use contain candid, first-person accounts of consequences that never appear in any clinical or administrative record.

This project asks a simple question: **Can we train a model to automatically identify those consequences from free-form social media text?**

Specifically, we draw inspiration from SMM4H-HeaRD 2026 Task 7, a Named Entity Recognition challenge where the model must extract two types of self-reported impacts of opioid use from Reddit posts — health consequences (withdrawal, overdose, depression) and social consequences (job loss, arrest, relationship breakdown). The published state-of-the-art achieves F1 = 0.61 with recall of just 0.52, against a human expert ceiling of Cohen's Kappa = 0.81 — a gap large enough that meaningful improvement is both possible and practically significant.

Two concrete problems explain most of this gap. First, the training set is Clinical-heavy (60% Clinical / 40% Social) while the test set is Social-heavy (70% Social / 30% Clinical), directly biasing the model toward the wrong entity type at test time. Second, many impacts are expressed in colloquial language rather than explicit medical or legal vocabulary — "I lost everything" or "those weeks were hell" — making them invisible to a model that has never been given a definition of what it is looking for.

We investigate five concrete questions:

- Does explicitly correcting for the mismatch between training and test entity distributions improve performance on the harder, underrepresented entity type?
- Does prepending natural language definitions of entity types to the model input improve recall on impacts expressed in colloquial rather than clinical language?
- Does multi-task training with entity presence detection as an auxiliary objective produce encoder representations that are more sensitive to entity-bearing contexts?

- Does augmenting training with synthetically generated examples targeting documented failure modes — implicit expressions, negated non-entities, and Social/Clinical boundary cases — improve recall beyond what the original dataset alone can achieve?
- Does training on examples ordered by difficulty (explicit entities first, implicit and ambiguous cases later) lead to better generalization compared to random sampling?

We tackle these questions by fine-tuning DeBERTa-large — the published baseline model — as well as GLiNER and a span-based NER model, with each intervention applied independently and then combined, allowing us to isolate the contribution of each.

Stretch goals (if time permits): task-specific instruction tuning of Llama-3-8b as a lightweight deployable alternative; ensemble distillation of all trained models into a single student; and submission to the Codabench leaderboard for external validation against other participating teams.

We believe this project is likely to succeed because every intervention is motivated directly by evidence in the data and in the baseline paper's own documented failure modes — not by architectural intuition. Prior work on analogous low-resource NER tasks has shown that targeted distributional corrections and definition-augmented inputs each produce reliable gains in exactly the settings we are working in.

2. NLP Task Description

The task is Named Entity Recognition (NER). Given a Reddit post written by someone describing their experience with substance use, the model reads the post token by token and labels each token as part of a health-related consequence, a social consequence, or neither.

Health consequences are called Clinical Impacts — things like withdrawal, overdose, or depression. Social consequences are called Social Impacts — things like losing a job, getting arrested, or a relationship breaking down. Everything else gets the label “O”.

The labelling follows the BIO scheme, where "B" marks the beginning of an entity span, "I" marks continuation, and "O" marks no entity:

Input/Output Example:

Input (Reddit post excerpt)	Expected BIO Output
"I am a recovering addict and I overdosed at 19 and I was charged with disorderly conduct"	["O", "O", "O", "O", "B-ClinicalImpacts", "O", "O", "B-ClinicalImpacts", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "O", "B-SocialImpacts", "I-SocialImpacts", "I-SocialImpacts", "I-SocialImpacts", "O", "O", "O", "O", "O", "O", "O", "O"]

Clinical Impacts: "addict" (B-Clinical Impacts), "overdosed" (B-Clinical Impacts)

Social Impacts: "charged" (B-Social Impacts), "with disorderly conduct" (I-Social Impacts ×3)

One constraint makes this harder than standard NER: only the impacts that the post author personally experienced count as entities. If someone writes "my friend lost his job", that is not annotated. The model must track who is experiencing the consequence, not just whether a consequence is mentioned.

A second difficulty is that many impacts are expressed without any clinical or legal vocabulary at all. "Those weeks were absolute hell" carries the same meaning as "I experienced severe withdrawal" but gives the model no explicit surface cue to work with. This is the core challenge the project is designed to address.

Performance is measured with **Relaxed F1** as the primary metric — partial overlap between a predicted span and the gold span of the correct type counts as partial credit. **Strict F1**, requiring exact boundary and type match, is reported as a secondary metric.

3. Data

Official Dataset

We use RedditImpacts 2.0, provided by the shared task organizers (Dey et al., PSB 2026). It contains 1,378 Reddit posts collected from opioid-related subreddits, annotated by two trained annotators following detailed guidelines refined by a domain expert. Inter-annotator agreement reached Cohen's Kappa = 0.81 before independent annotation began.

Split	Posts	SocialImpacts	ClinicalImpacts	Total Entities	Tokens
Train	842	408 (40%)	616 (60%)	1,024	17.2K
Dev	258	167 (43%)	223 (57%)	390	5.2K
Test	278	256 (70%)	108 (30%)	364	6.2K
Total	1,378	831 (47%)	947 (53%)	1778	28.6K

Preprocessing

We keep preprocessing minimal. Reddit language — slang, abbreviations, fragmented sentences — is part of what the model needs to understand, so aggressive cleaning would remove meaningful signal. We only replace usernames and URLs with placeholder tokens. We do not apply lowercasing. Tokenization is aligned to each model's own tokenizer using the first-subword strategy for multi-subword tokens.

For definition prompting, entity descriptions derived from the official annotation guidelines are prepended to each post before tokenization, separated by a [SEP] token. These definition tokens are masked from the training loss, so they inform the model without introducing spurious label targets.

Synthetic Data

The baseline paper showed that F1 plateaus at around 50% of the training data — meaning more of the same kind of data is unlikely to help. Therefore, rather than generating random paraphrases, we generate examples that specifically target the failure modes documented in the baseline paper:

- **Implicit expression posts:** impacts described without explicit medical or legal vocabulary, targeting the recall gap.
- **Negated non-entity posts:** posts where impact phrases are explicitly negated ("no criminal record") with correct O labels, teaching the model to handle negation.
- **Social/Clinical boundary posts:** ambiguous cases where the entity type requires contextual inference rather than vocabulary matching.
- **Social Impact-heavy posts:** to partially rebalance the training distribution toward the test set's 70/30 skew.

We use Claude and GPT-4o to generate these examples with structured prompts that specify the entity spans upfront. Every generated example is verified programmatically — span alignment is checked character by character, unnatural text is filtered using GPT-2 perplexity, and near-duplicates of real training examples are removed. We estimate retaining around 75% of generated samples after filtering.

The contribution of synthetic data is measured through ablation — we train with and without it and report the difference directly, so it is treated as a hypothesis to test rather than an assumption.

4. Neural Models

We evaluate four architecturally distinct approaches to the NER task, treating them as our baseline comparison before applying any innovations. This answers a preliminary question — which paradigm works best for this specific task — before we ask whether targeted interventions can push performance further.

Model 1 — BiLSTM-CRF

A classical sequence labelling model. Words are represented as GloVe embeddings, passed through a bidirectional LSTM to capture left and right context simultaneously, and decoded with a Conditional Random Field layer that enforces valid BIO tag transitions. We implement this ourselves in PyTorch from scratch. It represents the pre-transformer approach to NER and serves as a lower bound on what contextual representations alone can achieve without large-scale pretraining.

Model 2 — DeBERTa-large

We fine-tune DeBERTa-large from HuggingFace with a linear token classification head for BIO labelling. This is the architecture behind the published state-of-the-art F1 = 0.61 on this exact dataset. The model weights are downloaded from HuggingFace. The fine-tuning pipeline — data loading, loss computation, training loop, and evaluation — is implemented by us. Our reproduction of the published result is the controlled comparison point against which all innovations are measured.

Model 3 — GLiNER

This model is downloaded from HuggingFace (urchade/gliner_large-v2.1) and fine-tuned on our training data. GLiNER is architecturally designed to take natural language entity type descriptions as input — it detects spans by matching them against provided definitions rather than learning a fixed label vocabulary. This makes it a principled architectural comparison for Innovation 2 (definition prompting): where Innovation 2 adds definitions as a text prefix to DeBERTa, GLiNER integrates entity descriptions directly into its core matching mechanism. If GLiNER does not outperform our definition-prompted DeBERTa, that tells us something genuinely interesting — that a surface-level prompting approach can match a dedicated architecture built for the same purpose.

Model 4 — Span-Based NER

Rather than labelling token by token, this model enumerates all possible contiguous spans up to a maximum length and classifies each span directly as Clinical Impacts, Social Impacts, or neither. This sidesteps BIO decoding entirely — there are no invalid tag sequence errors possible because there is no tag sequence. The encoder is DeBERTa-large, with a span classification head on top that takes the first and last token representations of each candidate span as input. This architecture was used by the 2024 competition winner (UKYNLP, ACL 2024) and achieved F1 = 0.47 on the previous version of this task. We implement the span enumeration and classification head ourselves on top of the HuggingFace DeBERTa encoder.

Innovations Applied to the Best Baseline

After establishing which of the four models performs best, we apply our innovations to that model. We expect DeBERTa-large to be the strongest baseline based on the published results, but we confirm this empirically before proceeding. Each innovation is applied independently first, then in combination:

Innovation 1 — Distribution-Aware Training: Focal loss with higher weights on Social Impacts tokens, entity-balanced batch sampling, and a progressive reweighting schedule. This approach targets the train/test distribution mismatch and is implemented by us.

Innovation 2 — Entity Definition Prompting: Natural language definitions of both entity types prepended to each input before tokenization, derived from the official annotation guidelines. Definition tokens are masked from the training loss. Implemented by us as an input construction change — no architectural modification required.

Innovation 3 — Auxiliary Task Learning: Two additional classification heads on the shared encoder alongside the main NER head — one predicting entity presence per sentence (labels derived from BIO annotations), one predicting negation scope per token (labels auto-generated via NegEx). Multi-task training loop implemented by us.

Innovation 4 — Synthetic Data with Curriculum Learning: Synthetically generated examples targeting implicit expressions, negated non-entities, and Social/Clinical boundary cases, introduced gradually during training — easy, explicit entities first, hard and ambiguous cases later. Curriculum ordering and data generation pipeline implemented by us.

5. Baselines

Our four models serve as internal baselines against each other. We additionally evaluate our models against a combination of internal reference points and externally published scores to contextualize our results within the broader literature:

BiLSTM-CRF: Will be implemented by us. This model establishes the pre-transformer performance floor on this task and confirms our evaluation pipeline is working correctly before we move to larger models.

DeBERTa-large (published): It has a performance of F1 = 0.61, Precision = 0.75, and Recall = 0.52, as reported in Dey et al. (PSB 2026). This is the primary benchmark our innovations are measured against. We reproduce those metrics before claiming any improvement over it.

GLiNER (zero-shot): Before fine-tuning GLiNER on our training data, we evaluate it zero-shot using the entity type names as descriptions. This tells us how much of GLiNER's performance comes from its pretraining versus task-specific fine-tuning and gives a reference point for how definition-aware architectures perform without any supervision on this domain.

GPT-4o 3-shot: It has a performance of F1 = 0.44, as reported in Dey et al. (PSB 2026). We compare against the published score without re-running it. It represents the upper bound of what prompting a large language model without fine-tuning can achieve, and contextualizes how much fine-tuning on task-specific data is worth relative to pure model scale.

6. Evaluation

Quantitative Metrics (per shared task):

- Relaxed F1 (Primary): It gives partial credit when a predicted span overlaps with the correct gold span of the same entity type, rather than requiring exact boundary matches.
- Strict F1: it requires the predicted span to match the gold span exactly. The gap between Relaxed and Strict tells us how often the model is finding the right region but getting the edges wrong — a specific weakness the span-based model is designed to address.

Both metrics are reported three ways: overall, Clinical Impacts only, and Social Impacts only. Reporting per-entity is non-negotiable for this project because our central claim is that current models underperform on Social Impacts specifically. An improvement to overall F1 that comes entirely from Clinical would mean our interventions did not actually work.

What We Are Comparing Against

The main benchmark is F1 = 0.61 from Dey et al. (PSB 2026) — the official baseline on this dataset. Everything we do is measured against that number. Beyond that, we situate our results in the broader context of progress on this task:

Reference Point	Relaxed F1	What It Represents
GPT-4o 3-shot	0.44	Best LLM without fine-tuning
DeBERTa-large	0.61	Current state-of-the-art

We do not re-run the LLM experiments. We compare against their published scores directly.

Ablation Design

Each innovation is tested independently before being combined. Every row in the ablation table changes exactly one thing from the row above it, enabling precise attribution of any performance change:

Model	What Is Added
BiLSTM-CRF	Starting point
DeBERTa-large	Pretrained encoder
+ Distribution-Aware Training	Corrects train/test shift
+ Definition Prompting	Adds semantic guidance
+ Auxiliary Tasks	Shapes encoder representations
+ Synthetic Data + Curriculum	Targeted augmentation and ordering
Combined	Everything together
GLiNER	Architectural alternative
Span-Based NER	Architectural alternative

If the combined system beats every individual row, the interventions are complementary. If one row accounts for most of the gain, we know where the real contribution lies. Either outcome is a valid finding.

Qualitative Error Analysis

Numbers alone cannot tell us whether we solved the right problems. A model that improves overall F1 by getting better on easy, explicit entities while still missing implicit ones has not addressed what makes this task hard. Therefore, alongside the quantitative results, we manually examine a sample of errors on the test set, specifically looking at how often failures involve negated phrases, implicit expressions, third-person descriptions, and Social/Clinical boundary cases — the four error types the baseline paper documented. We compare this breakdown before and after our innovations to check whether the improvements are coming from the places we intended.

7. Timeline and Work Division

Week	Period	Tasks	Owner
1	Mar 1–7	Data EDA: verify entity distributions, span lengths, implicit vs explicit entity breakdown; preprocessing and tokenization alignment; set up training and evaluation pipeline	All
2	Mar 8–14	Implement BiLSTM-CRF baseline; fine-tune DeBERTa-large and reproduce published F1 = 0.61; establish evaluation script with per-entity Relaxed and Strict F1 and bootstrap CIs	All
3	Mar 15–21	Implement Innovation 1 (focal loss, entity-balanced sampling, progressive reweighting); run independent ablations for each component. Implement Innovation 2 (definition crafting from guidelines, input formatting, loss masking); run definition-only experiments	A+B / B+C
4	Mar 22–28	Implement Innovation 3 (auxiliary heads — entity presence and negation scope; auto-generate labels via NegEx); synthetic data generation pipeline with quality control; fine-tune GLiNER and span-based NER	All
5	Mar 29–Apr 4	Implement curriculum learning ordering; run all individual innovation ablations; combine all innovations into final system; tune combined approach on dev set	All
6	Apr 5–11	Full ablation table; per-entity error analysis stratified by failure mode; bootstrap CIs on all results; Codabench leaderboard submission; final report writing, results visualization, presentation preparation	All

8. References

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